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Project Name: Residential Power Electricity Supply (ES) Service Charges

Company Used: Virginia Electric and Power Company

https://cdn-dominionenergy-prd-001.azureedge.net/-/media/pdfs/virginia/residential-rates/schedule-1.pdf?la=en&rev=418fcae22c2c4dfca1d844f3d1c5690b&hash=24094908FF111143A649A4C43DB4DF35

Objectives:

(1.) Calculate the Electricity Supply (ES) Service Charges for the Generation kWh Charge of the residents of Virginia within each range of usage manually using the Arithmetic method.

(2.) Write a piecewise function of the service charges.

(3.) Recalculate the same ES Service Charges for the Generation kWh Charge for Virginia residents algebraically using the Piecewise Function method.

Information:

Electricity Supply (ES) Service Charge

Paragraph II.B. is not applicable to Customers receiving Electricity Supply Service from a Competitive Service Provider, except for non-bypassable charges in the

Exhibit of Applicable Riders, as discussed in Paragraph V., below:

Billing Months of June-September

First 800 ES kWh @ 2.9421¢ per KWh

Over 800 ES kWh @ 4.4768¢ per KWh

Conversions:

$$2.9421 \; cents \; per \; ES \; kWh = \frac{2.9421}{100} = \$0.029421 \; per \; ES \; KWh$$

$$4.4768 \; cents \; per \; ES \; kWh = \frac{4.4768}{100} = \$0.044768 \; per \; ES \; KWh$$

Arithmetic Method:

Amounts to Test are: 0 ES KWh, 700 ES KWh, 1000 ES KWh

(1.) 0 ES KWh

$$Cost = 0.029421(0)$$

$$Cost = \$0.00$$

(2.) 700 ES KWh

$$Cost = 0.029421(700)$$

$$Cost = \$20.5947$$

$$Cost \cong \$20.59$$

(3.) 1000 ES KWh

$$Cost = 0.029421(800) + 0.044768(1000 - 800)$$

$$Cost = 23.5368 + 8.9536$$

$$Cost = 32.4904$$

$$Cost \cong \$32.49$$

Piecewise Function:

Define variables to be used:

Let *p* be the quantity of power consumed in ES KWh.

Let *c* be the cost per unit of power consumed in $.

First Piece:

$0 \leq p \leq 800$ ES KWh: $0.029421 per ES KWh

$$c(p) = 0.029421p$$

Second Piece:

$p > 800$ ES KWh: $0.04768 per ES KWh

$$c(p) = 0.029421(800) + 0.044768(p - 800)$$

$$c(p) = 23.5368 + 0.044768p - 35.8144$$

$$c(p) = 0.044768p - 12.2776$$

$$c(p) = \begin{cases} 0.029421p; & 0 \leq p \leq 800 \\ 0.044768p - 12.276 & p > 800 \end{cases}$$

Piecewise Function Method:

Amounts to Test are: 0 ES KWh, 700 ES KWh, 1000 ES KWh

(1.) 0 ES KWh

$$c(p) = 0.029421(p)$$

$$c(p) = 0.029421(0)$$

$$c(p) = \$0.00$$

(2.) 700 ES KWh

$$c(p) = 0.029421(p)$$

$$c(700) = 0.029421(700)$$

$$c(700) = 20.5947$$

$$c(700) \cong \$20.59$$

(3.) 1000 ES KWh

$$c(p) = 0.044768p - 12.276$$

$$c(1000) = 0.044768(1000) - 12.276$$

$$c(1000) = 44.768 - 12.2776$$

$$c(1000) = 32.4904$$

$$c(1000) \cong \$32.4$$

References (APA 7):

Virginia Electric and Power Company. (n.d.). Retrieved from https://cdn-dominionenergy-prd-001.azureedge.net/-/media/pdfs/virginia/residential-rates/schedule-1.pdf?la=en&rev=418fcae22c2c4dfca1d844f3d1c5690b&hash=24094908FF111143A649A4C43DB4DF35

Piecewise Functions. (n.d.). Precalculus.appspot.com. Retrieved on April 22, 2024 from https://precalculus.appspot.com/PiecewiseFunctions/piecewise-functions.html#studentProjectPowerBill